

WASP-2b

R-2432

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_110609 · created 2026-08-22T00:52:54+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2455721.9021 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.16 +/- 0.045
Transit depth (Rp/Rs)^2	2.55 +/- 1.45 [%]
Orbital Inclination (inc)	89.48 +/- 2.83
Airmass coefficient 1 (a1)	0.44 +/- 0.29
Airmass coefficient 2 (a2)	0.42 +/- 1.12
Scatter in the residuals of the lightcurve fit is	70.75 %
Best Comparison Star	#1 - [156, 454]
Optimal Aperture	0.87
Optimal Annulus	6.75
Transit Duration (day)	0.093 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.16 ± 0.045 vs 0.1326 (deviation 0.61σ)
Duration fit vs published	0.093 d vs 0.0737 d (deviation 1.6σ)
Mid-time O–C	0.3 min
Depth SNR	3.6
Residual scatter	70.75 %

Light curve

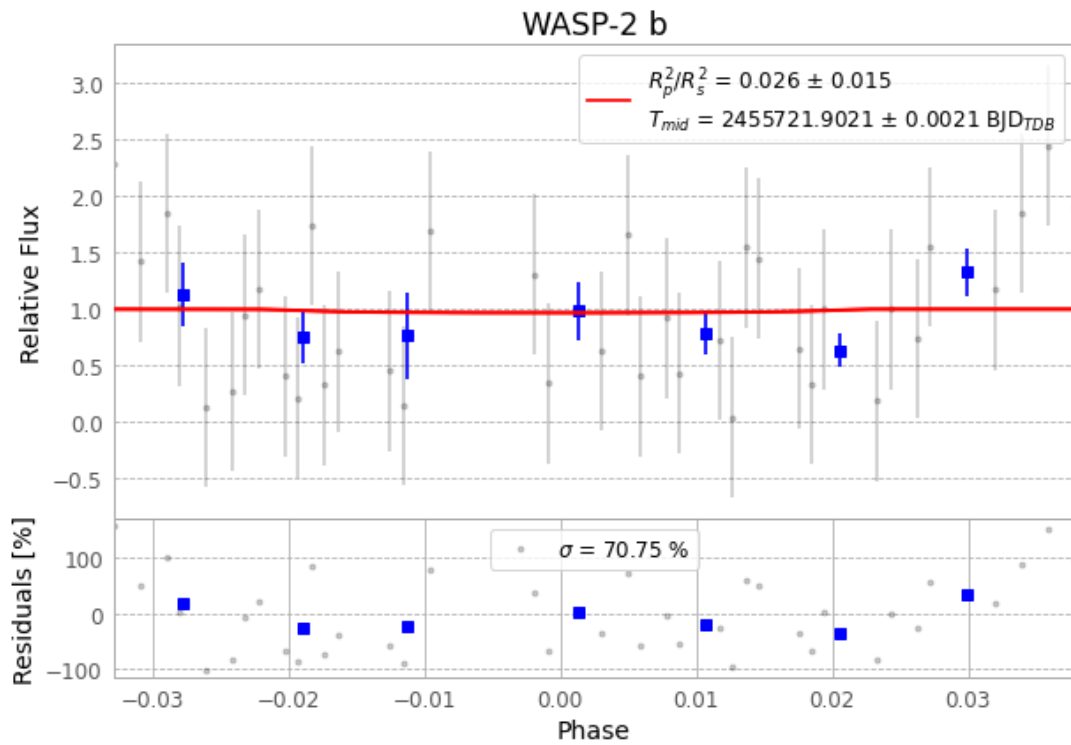


Figure 1 — FinalLightCurve_WASP-2 b_2011-06-09.png (SHA-256 f15b762f612c7719...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [101, 481], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bf37837c8d75b368...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

76 FITS frames (50 MB), WASP-2110609075113.FITS → WASP-2110609113613.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-110609.log (068660aed2e7...), FinalLightCurve_WASP-2 b_2011-06-09.png (f15b762f612c...), FinalLightCurve_WASP-2 b_2011-06-09.csv (0ab508141d82...)

Compute resources

Wall time 123.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-22 00:52Z.